ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
GUWAHATI

Course Structure and Syllabus
(From Academic Session 2018-19 onwards)

B.TECH
ELECTRICAL ENGINEERING
8th SEMESTER

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure

(From Academic Session 2018-19 onwards)

B. Tech 8th Semester: Electrical Engineering

Semester VIII/ B. TECH/EE

Sl. No.	Sub-Code	Subject	Hours per Week			Credits	Marks	
			L	T	CE	CE	CE	ESE
Theory								
1	EE1818PE5*	Program Elective-5	3	0	0	3	30	70
2	EE1818PE6*	Program Elective-6	3	0	0	3	30	70
3	EE1818PE7*	Program Elective-7	3	0	0	3	30	70
4	EE1818OE4*	Open Elective-4	3	0	0	3	30	70
Practical								
1	EE181821	Project -2	0	0	12	6	100	50
TOTAL			**12**	**0**	**12**	**18**	**220**	**330**
Total Contact Hours per week : 24								
Total Credits: 18								

Program Elective-5 Subjects		
Sl. No.	**Subject Code**	**Subject**
1	EE1818PE51	Electrical Engineering Professional Practices
2	EE1818PE52	Smart Grid Technology
3	EE1818PE53	VLSI Signal processing
4	EE1818PE54	Microwave Theory and Techniques
5	EE1818PE55	HVDC
6	EE1818PE5*	Any other subject offered from time to time with the approval of the University

Program Elective-6 Subjects		
Sl. No.	**Subject Code**	**Subject**
1	EE1818PE61	Power System Dynamics and Control
2	EE1818PE62	Illumination Engineering
3	EE1818PE63	Power Quality Analysis and Assessment
4	EE1818PE64	Industrial Electrical Systems
5	EE1818PE6*	Any other subject offered from time to time with the approval of the University

Program Elective- 7 Subjects		
Sl. No.	**Subject Code**	**Subject**
1	EE1818PE71	Utilization and Conservation of Electrical Energy
2	EE1818PE72	Artificial Intelligence Applications in Electrical Engineering
3	EE1818PE73	Power System Instrumentation
4	EE1818PE7*	Any other subject offered from time to time with the approval of the University

Open Elective-4 Subjects		
Sl. No.	**Subject Code**	**Subject**
1	EE1818OE41	Reliability Engineering
2	EE1818OE42	Web Technology
3	EE1818OE43	Optical Communication Engineering
4	EE1818OE44	Computer Graphics
5	EE1818OE4*	Any other subject offered from time to time with the approval of the University

Detail Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE51	Electrical Engineering Professional Practices	3-0-0	3

COURSE OBJECTIVES:
The main objective of this course is to introduce the students to the basic concepts of installation, testing, commissioning of electrical machines, estimation & costing of electrical machines and wiring, and preparation of tender papers.

COURSE OUTCOMES: After completion of the course students will be able to:

CO1: describe procedures of installation, testing, commissioning of electrical machines, estimation & costing of electrical machines and wiring, and preparation of tender papers

CO2: solve problems of installation, testing, commissioning of electrical machines, estimation & costing of electrical machines and wiring, and preparation of tender papers.

CO3: assess installation, testing, commissioning of electrical machines, estimation & costing of electrical machines and wiring, and tender papers.

MODULE 1: Transformers

Installation: Location & site; Foundation; code of practice for terminal plates; polarity & phase sequence; Oil tanks; Drying of windings with & without oil; General inspection.

Commissioning Tests: Voltage ratio; Earthing resistance; Oil strength; Buchholz & other relays; Tap changing gear; Fans & pumps: Insulation tests; Impulse test; Load & temperature rise tests

Specific Tests: Determination of performance & efficiency

MODULE 2: Induction Motors

Installation: Location of motor & control apparatus; Foundation & leveling; Shaft alignment for various coupling; Drying out of windings.

Commissioning Tests: Mechanical tests for alignment; Air-gap symmetry; bearings; Vibrations & balancing.

Electrical tests- Insulation test; Earthing; High voltage test; Starting up; Ability to speed up and take load.

Type & Routine Tests: In accordance with ISI Test code

Specific Tests: Performance & temperature rise tests; stray load losses; shaft currents

Maintenance Schedule

MODULE 3: Synchronous Machines

Installation: Physical inspection & alignment check; excitation system; cooling & control gear; Drying out.

Commissioning Tests: Insulation resistance; Resistance of armature & field windings; Polarity & phase sequence; Shaft current; Waveform & telephone interference factor; over speed test; Line charging capacity.

Performance Tests: Various tests to estimate the performance for generator and motor operation; Sudden short circuit tests and transient and sub-transient parameters; measurement of sequence impedances; separation of losses; Retardation test; temperature rise test.

Factory Tests: Gap length; Magnetic centrity; Balancing; Vibration; Bearing currents; Electrical tests.

MODULE 4: Estimating and Costing Electrical Wiring
Estimation of electrical load, size and type of electrical conductors, MCBs, Panels, TPNs, switches and sockets, Ceiling roses, Air conditioner points, Fan and Light points

MODULE 5: Tender Documents
Preparation and assessment of tender papers

TEXT BOOKS:
1. Electrical Wiring Estimating & Costing 2008 – S. L. Uppal, Khanna Publishers.
2. Installation, Commissioning & Maintenance of Electrical Equipment – P. P. Gupta, Dhanpat Rai Publications
3.

REFERENCE BOOKS:
1. Elect. Installation Works 6th Ed 1996 – T. G. Francis and R. J. Cooksley, Longman Higher Education, ELBS.
2. Electrical Installation and Workshop Technology (Vol-1 & 2) 4th Ed 1992 - F. G. Thompson, Longman Higher Education, ELBS
3. Installation, Commissioning & Maintenance of Electrical Equipment 2013 - Tarlok Singh, S. K. Kataria & Sons
4. Testing, Commissioning, Operation & Maintenance Of Electrical Equipments 2016 – S. Rao
5. A Course in Electrical Installation Estimating and Costing 2013 – J. B. Gupta, S. K. Kataria & Sons
6. Design & Testing of Elect. Machines - M. V. Deshpande, Wheeler
7. Electrical Substation Engineering & Practice: EHV-HVDC & SF-GIS (Principle, Practice, Design and Reference Data) 2003 – Sumathi Rao, Khanna Publishers
8. Electrical Workshop: Safety, Commissioning, Maintenance & Testing of Electrical Equipment 3rd Ed 2012 – R. P. Singh, I K International Publishing House Pvt. Ltd

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE52	**Smart Grid Technology**	**3-0-0**	**3**

COURSE OBJECTIVES:
To study distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid

COURSE OUTCOMES (COs): After completion of the course students will be able to
CO1: describe structure, components and mathematical formulations for distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid
CO2: solve problems of distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid
CO3: analyze performance and characteristics of distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid

MODULE 1: Introduction to Smart Grids
Definition, justification for smart grids, smart grid conceptual model, smart grid architectures, Interoperability, communication technologies, role of smart grids standards, intelligrid initiative, national smart grid mission (NSGM) by Govt. of India

MODULE 2: Smart Transmission Technologies
Substation automation, Supervisory control and data acquisition (SCADA), energy management system (EMS), phasor measurement units (PMU), Wide area measurement systems (WAMS)

MODULE 3: Smart Distribution Technologies
Distribution automation, outage management systems, automated meter reading (AMR), automated metering infrastructure (AMI), fault location isolation and service restoration (FLISR), Outage Management Systems (OMS), Energy Storage, Renewable Integration

MODULE 4: Distributed Generation and Smart Consumption
Distributed energy resources (DERs), smart appliances, low voltage DC (LVDC) distribution in homes or commercial buildings, home energy management system (HEMS), Net Metering, Building to Grid B2G, Vehicle to Grid V2G, Solar to Grid, Microgrid

MODULE 5: Regulations and Market Models for Smart Grid
Demand Response, Tariff Design, Time of the day pricing (TOD), Time of use pricing (TOU), Consumer privacy and data protection, consumer engagement; Cost benefit analysis of smart grid projects

TEXT BOOKS:
1. Clark W. Gellings: The Smart Grid, Enabling Energy Efficiency and Demand Side Response, CRC Press, 2009.
2. Jean Claude Sabonnadière, Nouredine Hadjsaïd: Smart Grids, Wiley-ISTE, IEEE Press, May 2012
REFERENCES:
1. Janaka Ekanayake, Kithsiri Liyanage, Jianzhong. Wu, Akihiko Yokoyama, Nick Jenkins: Smart Grid: Technology and Applications, Wiley, 2012.
2. James Momoh: Smart Grid: Fundamentals of Design and Analysis, Wiley, IEEE Press, 2012.
3. India Smart Grid Knowledge Portal

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE53	**VLSI Signal processing**	**3-0-0**	**3**

COURSE OBJECTIVES:

- To teach Advance Concepts of Digital Signal Processing Algorithms

COURSE OUTCOMES: At the end of this course, the students will be able to

CO1: describe structure, components and mathematical formulation of VLSI signal processing systems
CO2: analyze VLSI signal processing systems
CO3: design VLSI signal processing systems for utilization in societal, academic and industrial purposes

MODULE 1: Iteration Bound
Typical DSP algorithms, DSP application demands and scaled CMOS technology, Representation of DSP algorithms; Data-flow graph representations, Loop bound and iteration bound, Algorithms for computing iteration bound, Iteration bound of multirate data-flow graphs

MODULE 2: Pipelining and Parallel Processing
Pipelining of FIR digital filters, Parallel processing, Pipelining and parallel processing for low power; Numerical strength reduction – Synchronous, Wave and Asynchronous pipe lines; Low power design

MODULE 3: Retiming, Unfolding, Folding
Retiming: Definitions and properties, Solving systems of inequalities, Retiming techniques.

Unfolding: An algorithm for unfolding, Properties of unfolding, Critical path, unfolding and retiming, Applications of unfolding.

Folding: Folding transformation, Register minimization techniques, Register minimization in folding architectures, Folding of multirate systems

MODULE 4: Systolic Architecture Design
Systolic array design methodology, FIR systolic arrays, Selection of scheduling vector, Matrix-matrix multiplication and 2D systolic array design, Systolic design for space representations containing delays; Scaling and round off noise; Digital lattice filter structures

MODULE 5: Bit-Level Arithmetic Architecture
Parallel multipliers, Interleaved floor-plan and bit-plane-based digital filters, Bit-serial multipliers, Bit-serial filter design and implementation, Canonic signed digit arithmetic, Distributed arithmetic; Bit level arithmetic architecture; Redundant arithmetic

MODULE 6: Programmable Digital Signal Processors
Evolution of programmable digital signal processors, Important features of DSP processors, DSP processors for mobile and wireless communications, Processors for multimedia signal processing

TEXT BOOKS:

1. K. K. Parhi: VLSI Digital Signal Processing Systems, Design and Implementation, Wiley Interscience, New Delhi, 1999

REFERENCES:

1. K. P. Keshab: VLSI Digital Signal Processing Systems: Design and Implementation, Jacaranda Wiley, 1999

2. Richard J. Higgins: Digital Signal Processing in VLSI , Prentice Hall
3. M.A. Bayoumi: VLSI Design Methodology for DSP Architectures, Kluwer, 1994
4. Mohammad Ismail and Terri Fiez: Analog VLSI signal and information processing, McGraw-Hill
5. S.Y. Kung, H.J. White House, T. Kailath: VLSI and Modern Signal Processing, Prentice Hall, 1985
6. Multirate Systems and Filter Banks 1st Ed 1992 – P. P. Vaidyanathan, Prentice Hall
7. Digital Signal Processing: A Practical Approach 2nd Ed 2004 - Emmanuel Ifeachor and Barrie Jervis, Prentice Hall
8. Discrete-Time Signal Processing 2nd Ed 1999 – Alan V. Oppenheim, Ronald W. Schafer, John R. Buck, PHI
9. Adaptive Signal Processing 1985 - Bernard Widrow and Peter N. Stearns, Pearson
10. Adaptive Filter Theory 5th Ed 2013 - Simon Haykin, Pearson
11. Wavelets and Filter Banks 2nd Ed 1996 - G Strang and T Nguyen, Wellesley-Cambridge Press
12. Time Frequency analysis: Theory and Applications 1994 - Leon Cohen, Prentice Hall
13. Optimum Signal Processing: An Introduction 2nd Ed 1988 - Sophocles J Orfanidis, Collier Macmillan
14. J. G. Proakis, D.G. Manolakis, "Digital Signal Processing: Principles, Algorithms And Applications", Pearson 2007.
15. Digital Signal Processing 2014 - Tarun Kumar Rawat, Oxford University Press

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE54	**Microwave Theory and Techniques**	**3-0-0**	**3**

COURSE OBJECTIVES:
The main objective of this course is to introduce the students to the basic concepts of telecommunication through Microwave wireless devices and systems

COURSE OUTCOMES (COs): After completion of the course students will be able to:
CO1: Explain microwave system components their properties
CO2: Analyze microwave systems
CO3: Design microwave systems for practical application

MODULE 1: History and Applications
History of Microwaves, Microwave Frequency bands; Applications of Microwaves: Civil and Military, Medical, EMI/ EMC

MODULE 2: Mathematical Model of Microwave Transmission
Concept of Mode, Features of TEM, TE and TM Modes, Losses associated with microwave transmission, Concept of Impedance in Microwave transmission

MODULE 3: Analysis of RF and Microwave Transmission Lines
Coaxial line, Rectangular waveguide, Circular waveguide, Strip line, Micro strip line

MODULE 4: Microwave Network Analysis
Equivalent voltages and currents for non-TEM lines, Network parameters for microwave circuits, scattering Parameters

MODULE 5: Passive and Active Microwave Devices
Passive Microwave Devices: Directional Coupler, Power Divider, Magic Tee, Attenuator, Resonator
Passive Microwave Devices: Diodes, Transistors, Oscillators, Mixers; Microwave Semiconductor Devices: Gunn Diodes, IMPATT diodes, Schottky Barrier diodes, PIN diodes; Microwave Tubes: Klystron, TWT, Magnetron

MODULE 6: Microwave Design Principles
Impedance transformation, Impedance Matching, Microwave Filter Design, RF and Microwave Amplifier Design, Microwave Power Amplifier Design, Low Noise Amplifier Design, Microwave Mixer Design, Microwave Oscillator Design

MODULE 7: Microwave Antennas
Antenna parameters, Antenna for ground based systems, Antennas for airborne and satellite borne systems, Planar Antennas

MODULE 8: Microwave Measurements
Power, Frequency and impedance measurement at microwave frequency, Network Analyzer and measurement of scattering parameters, Spectrum Analyzer and measurement of spectrum of a microwave signal, Noise at microwave frequency and measurement of noise figure; Measurement of Microwave antenna parameters

MODULE 9: Microwave Systems
Radar, Terrestrial and Satellite Communication, Radio Aids to Navigation, RFID, GPS

MODULE 10: Modern Trends in Microwaves Engineering
Effect of Microwaves on human body, Medical and Civil applications of microwaves, Electromagnetic interference and Electromagnetic Compatibility (EMI & EMC), Monolithic Microwave ICs, RFMEMS for microwave components, Microwave Imaging

TEXT BOOKS:
1. R.E. Collins: Microwave Circuits, McGraw Hill
2. A. K. Maini: Microwaves and Radar – Principles and Applications‖, Khanna Publishers

REFERENCE BOOKS:
1. K.C. Gupta, I.J. Bahl: Microwave Circuits, Artech house
2. V. Chandrasekar: Communication System, Oxford University press, India

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE55	**HVDC**	**3-0-0**	**3**

Objectives: Introduce fundamental concepts of High Voltage DC Transmission system
Course Outcomes: After the completion of this course the student will be able to
CO1: explain working principles and protection of HVDC devices and systems
CO2: design HVDC system
CO3: analyze characteristics and performances of HVDC devices and systems

MODULE 1: DC Transmission Technology
Comparison of AC and DC transmission; Application of DC transmission; Description of DC transmission system planning for HVDC transmission; Modern trends in DC transmission.

MODULE 2: Analysis of HVDC Converters
Pulse number; Choice of converter configuration; Simplified analysis of Graetz circuit; Line Commutated Converters (LCCs): Six pulse converter; Analysis neglecting commutation overlap, harmonics, Twelve Pulse Converters; Inverter Operation; Effect of Commutation Overlap; Expressions for average dc voltage, AC current and reactive power absorbed by the converters; Effect of Commutation Failure, Misfire and Current Extinction in LCC links

MODULE 3: Converter and HVDC System Control
General principles of DC link control; Converter control characteristics; System control hierarchy; Firing angle control– Phase-Locked Loop; Current and Extinction angle control; Starting and Stopping of DC link; Power control; Higher level controllers; Telecommunication requirements

MODULE 4: Harmonics and Filters
Introduction-generation of harmonics; Design of AC filters; DC filters; Carrier frequency and Radio Interference (RI) noise

MODULE 5: Stability Enhancement Control
Basic Concepts: Power System Angular, Voltage and Frequency Stability; Power Modulation: basic principles – synchronous and asynchronous links; transient stability improvement using DC link.

Textbooks/ Reference Books:

1. K. R. Padiyar: HVDC power transmission system, Wiley Eastern Limited, New Delhi, 1990.
2. P. Kundur: Power System Stability and Control, Tata McGraw-Hill Publishing Company Ltd., USA, 1994.
3. J. Arrillaga: High Voltage Direct Current Transmission, Peter Pregrinus, London, 1983.
4. Edward Wilson Kimbark: Direct Current Transmission, Vol. I, Wiley Interscience, New York, London, Sydney, 1971.
5. Rakosh Das Begamudre: Extra High Voltage AC Transmission Engineering, New Age International (P) Ltd., New Delhi, 1990.
6. S. S. Rao: EHV-AC, HVDC Transmission and Distribution Engineering, Khanna Publishers
7. S. L. Uppal and S. S. Rao: Electrical Power Systems, Khanna Publishers

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE61	**Power System Dynamics and Control**	**3-0-0**	**3**

MODULE 1: Power System Stability
Introduction power system dynamics – slow and fast dynamics, classifications and issues for applications. Modeling of a power system for dynamic analysis – lumped representation of a power system with single machine and single load concept, swing equation. Power- angle curve. Steady- state and transient Stabilities. Equal area criterion. Analysis of stability analysis using HVDC link in a power system. Calculation of power – angle curves for fault and post – fault conditions for various types of Fault; effect of reclosing. Numerical solution of swing equation.

MODULE 2: Synchronous Machine Modeling and Power System Dynamics
Fundamental of single and three phase synchronous machine, modeling of three-phase machine – voltage equation, induction matrix elements, Park's Transformation, Equivalent circuit. Operation of Three-phase synchronous machine under balanced steady-state condition, Transient and Sub-transient parameters of Three-phase synchronous machine and machine modeling with these parameters.

MODULE 3: Load Frequency Control (LFC) of a Power System
Introduction power system LFC Modeling of a power system for LFC as single area system with governor and turbine model. LFC model for two or multi-area power system. State model for LFC of single and multi-area power system. Fundamental of Optimal control, Riccati equation, Optimal LFC of a power system. Frequency control issues with generator rate control, speed governor dead-band effect.

MODULE 4: Small- Signal Stability Analysis of Power System
Generation representation using classical model, State equations and block diagram, State variable representation of generator voltage and electrical torque. Effect of excitation system Effect of AVR on synchronous a damping torque.

MODULE 5: Transient Stability Analysis for Multi-Machine Power System
Introduction. Modeling of interconnecting power system for multi-machine power system transient stability analysis. Modification of load flow analysis model for transient stability analysis, solution algorithm for transient stability analysis of an interconnected power system.

Textbooks/ Reference Books:

1. Computer Methods in Power System Analysis – Stagg and El-Abiad, McGraw-Hill publication
2. Power system interconnection and control - Murty.
3. Power System Dynamics Analysis and Simulation – R Ramanujam, PHI Learning PLTD

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE62	**Illumination Engineering**	**3-0-0**	**3**

COURSE OBJECTIVES:

- To impart the basic knowledge about radiation, entities in the illumination system, human eye and vision
- To inculcate the understanding of various mathematical relations between brightness and luminosity, light sources and their characteristics etc
- To understand light control
- To know about measurement of illumination and design.

COURSE OUTCOMES:

At the end of the course, the students will be able to:

CO1: Understand the physics behind various luminous sources, laws of illumination and illumination standards.

CO2: Assess the structure and visual characteristics of human eye with respect to response to glare and colors.

CO3: Evaluate the performance of various light sources based on their optical characteristics.

CO4: Compare illumination levels from various sources and perform measurement experiments for appropriate applications.

CO5: Design efficient lighting systems for various application areas.

MODULE 1: Radiation

Wavelength, frequency & velocity. The radiation spectrum. Radiations from black bodies & other sources.

MODULE 2: Entities in the Illumination System and their Units

Luminous sources, illumination, intensity, brightness, other terms and units. The inverse square law. The cosine law. Solid angle relationship. Luminosity, relationship between brightness & luminosity for a perfectly diffusing source, illumination standards.

MODULE 3: The Eye & Vision

The structure of the eye, accommodation, aberration of the eye, the rods & cones, visual acuity, glare, color & color response of the eye.

MODULE 4: Light Sources & Their Characteristics

Day light incandescent, electric discharge (low & high pressure), fluorescent, are amps and laser beams, color rendering, wiring, switching & control circuits. Starters & ballast.

MODULE 5: Light Control

Reflection & reflection factor, absorption, transmission & transmission factor. Control of light by luminaries.

MODULE 6: Illumination & Measurement

Illumination from point sources, light units in a row, area illumination, polar curves. Linear & surface sources, flat linear source, flat-strip of short length. Illumination of a vertical source. Radiant energy detectors, PV cell, Photo-tubes, Photometry, Electro-photometry, Photo cells, Spectro-photometer, Colorimeters.

MODULE 7: General Illumination & Design calculation
Interior lighting of industrial, residential & commercial buildings. Effective utilizations of daylight. Daylight factor, Outdoor lighting, Street, rail/shipyards, airports, sports area. Lighting design for signaling, advertising & security.

Textbooks/ Reference Books:

1. Cotton.H, Principles of Illumination, Chapman & Hall.
2. Boast. W.S., Illumination Engineering, McGraw-Hill.
3. IES Lighting Handbook, Illumination Engineering Society, New York.
4. Energy Management in Illuminating Systems—Kao Chen

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE63	**Power Quality Analysis and Assessment**	**3-0-0**	**3**

MODULE 1: Introduction

Power quality and its importance, transients, long duration voltage variations, short durations, voltage variations, waveform distortions, voltage fluctuations, power frequency variations.

MODULE 2: Voltage Sags and Interruptions

Sources of sags and interruptions, Estimating voltage sag performance, mitigation of voltage sags: active series compensators, Static transfer switches and fast transfer switches, solution at the end user level and their economic implications.

MODULE 3: Transients Overvoltage

Source of transient's overvoltage, principles of overvoltage protection, devices for overvoltage protection

MODULE 4: Harmonics

Harmonic distortions, sources of harmonics and their locations, effects of harmonics Harmonic distortion evaluation - devices for controlling harmonic distortion - passive and active filters. IEEE standards.

MODULE 5: Long Duration Voltage Variations

principles of regulating the voltage, devices for voltage regulation

MODULE 6: Power Quality Benchmarking

Benchmarking processes, RMS voltage variation indices, harmonic indices, Power Quality State Estimator (PQSE)

MODULE 7: Power Quality Monitoring

Monitoring considerations - monitoring and diagnostic techniques for various power quality problems - modeling of power quality (harmonics and voltage sag) problems by mathematical simulation tools - power line disturbance analyzer – Quality measurement, equipment - harmonic / spectrum analyzer - flicker meters - disturbance analyzer, Applications of expert systems for power quality monitoring

Textbooks/ Reference Books:

1. Roger. C. Dugan, Mark. F. Mc Granagham, Surya Santoso, H.Wayne Beaty, 'Electrical Power Systems Quality' McGraw Hill,2003.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE64	**Industrial Electrical Systems**	**3-0-0**	**3**

Course Outcomes: At the end of this course the student will be able to:

CO1: Explain the electrical wiring systems for residential, commercial and industrial consumers, representing the systems with standard symbols and drawings

CO2: Explain the construction and working components of industrial electrical systems.

CO3: Design the proper size of electrical system components.

MODULE 1: Electrical System Components

LT system wiring components, selection of cables, wires, switches, distribution box, metering system, Tariff structure, protection components- Fuse, MCB, MCCB, ELCB, inverse current characteristics, symbols, single line diagram (SLD) of a wiring system, Contactor, Isolator, Relays, MPCB, Electric shock and Electrical safety practices

MODULE 2: Residential and Commercial Electrical Systems

Types of residential and commercial wiring systems, general rules and guidelines for installation, load calculation and sizing of wire, rating of main switch, distribution board and protection devices, earthing system calculations, requirements of commercial installation, deciding lighting scheme and number of lamps, earthing of commercial installation, selection and sizing of components.

MODULE 3: Illumination Systems

Understanding various terms regarding light, lumen, intensity, candle power, lamp efficiency, specific consumption, glare, space to height ratio, waste light factor, depreciation factor, various illumination schemes, Incandescent lamps and modern luminaries like CFL, LED and their operation, energy saving in illumination systems, design of a lighting scheme for a residential and commercial premises, flood lighting.

MODULE 4: Industrial Electrical Systems-I

High Tension (HT) connection, industrial substation, Transformer selection, Industrial loads, motors, starting of motors, Single Line Diagram (SLD), Cable and Switchgear selection, Lightning Protection, Earthing design, Power factor correction – kVAR calculations, type of compensation, Introduction to PCC, MCC panels; Specifications of Low Tension (LT) Breakers, Miniature Circuit Breaker (MCB) and other LT panel components.

MODULE 5: Industrial Electrical Systems-II

Diesel Generator (DG) Systems, UPS System, Electrical Systems for the elevators, Battery banks, Sizing the DG, Uninterrupted Power Supply (UPS) and Battery Banks, Selection of UPS and Battery Banks

MODULE 6: Industrial Electrical System Automation

Study of basic Programmable Logic Controller (PLC), Role of in automation, advantages of process automation, PLC based control system design, Panel Metering and Introduction to SCADA system for distribution automation.

Text Books:

1. S. L. Uppal and G. C. Garg, ―Electrical Wiring, Estimating & Costing‖, Khanna publishers, 2008.
2. H. Joshi, ―Residential, Commercial and Industrial Systems‖, McGraw-Hill, 2008

References:

1. K. B. Raina, ―Electrical Design, Estimating & Costing‖, New age International, 2007
2. S. Singh and R. D. Singh, ―Electrical estimating and costing‖, Dhanpat Rai and Co., 1997
3. Web site for IS Standards.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE71	**Utilization and Conservation of Electrical Energy**	**3-0-0**	**3**

Course Outcomes:

At the end of the course, the students will be able to:

CO1: Analyze various procedures, circuits, practices employed for different types of electric heating.

CO2: Compare various types of welding, AC and DC power supplies used for welding and to choose circuits used in welding.

CO3: Justify the use of various traction motors in electric traction, apply their starting & speed control techniques and summarise knowledge of various energy storage devices.

CO4: To assess electrical energy losses and examine procedures for energy conservation and co-generation.

CO5: To associate with energy audit, its classification, methodology and instrumentation.

MODULE 1: Electric Heating

Advantages, Classification, Resistance Heating, Furnaces, Requirements and Design of heating elements, Temperature control, Electric arc furnaces, Direct & Indirect, Construction & Operation, Electrodes & Power Supply, High Frequency Heating, Induction Heating, Working principle, Power & High frequency Heating, Choice of Frequency, Core type & Coreless Furnaces, Skin Effect & Pinch effect, High Frequency Supply, Advantages & Disadvantages, Dielectric Heating, Working principle, Choice of Voltage and Frequency, Advantages & Applications.

MODULE 2: Electric Welding

Classifications, Resistance Welding: Spot, Butt, Seam. Arc welding: types, electrode used, power sources and control circuits. Atomic hydrogen welding. Modern development.

MODULE 3: Electric Traction

Advantages. Systems of electric traction. Choice of system voltage and frequency. The Indian scenario. Types of train services. Train movements and energy consumption. Speed-time, distance-time and energy consumption curves.Tractive effort, Adhesion, Train resistance. Power supply arrangements. Substation equipment. D.C AND A.C. traction motors, their disposition and operation on tram cars, motor coaches and locomotives. Control systems; Rheostatic, field control and seriesparallel using shunt and bridge transition methods. Multiple unit control. Metadyne control. Controllers for dc & ac traction motors. Tram Cars, Motor coaches, &Trolly Buses. Auxiliary Electrical Equipments for Tram cars, Motor Coaches & Locomotives. Braking: mechanical, vacuum & electrical.

MODULE 4: Energy Storage

Size & Duration of storage. Modes of energy storage: mechanical, electrical, magnetic, thermal & chemical. Comparison of the different systems.

MODULE 5: Electrical Losses & Energy Conversion

Electrical transmission, distribution & utilization losses. Classification. Reduction of losses. Benefits of electrical energy conservation. Energy conservation in lighting, electric furnaces, electric drive, traction systems. Use of energy –efficient equipment.

MODULE 6: Electrical Energy Audit

Introduction, benefits, procedure for energy audit. Instruments for energy audit. Methodology. Case study.

Textbooks/ Reference Books:

1. Electrical Energy Utilization & Conservation, by Tripathy, S.C ; TMG
2. Utilization of Electric power ; by Suryanarayan, N.V. : Wiley Eastern Ltd.
3. Art and Science of Utilization of Electrical Energy'by H Partab: Dhanpat Rai and Sons
4. Utilization of Electric Power and Electric Traction By J B Gupta: S K Kataria and Sons
5. Generation, Distribution and Utilization of Electrical Energy By C L wahhwa: New Age International (P) limited Publishers

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE72	**Artificial Intelligence Applications in Electrical Engineering**	**3-0-0**	**3**

COURSE OBJECTIVES:

- To provide fundamentals of intelligent systems technologies including knowledge-based systems, neural networks, fuzzy systems, and evolutionary computation.
- To provide adequate knowledge of application of Neural Networks, Fuzzy logic and hybrid model for solving Electrical Engineering problems.

COURSE OUTCOMES:

At the end of this course, the students will be able to:

CO1: Explain neural networks behavior and use them for classification, control system and optimization problem

CO2: Gain a working knowledge of fuzzy controllers used in engineering problems.

CO3: Obtain the optimum solution of well formulated optimization problem using evolutionary approaches.

CO4: Formulate hybrid intelligent algorithms for the application of engineering problems.

MODULE 1: Introduction

Definition, History of AI, Approaches to AI, AI technologies. Importance of AI, Application areas of AI, Comparisons with conventional program-classical problem solving methods and heuristic search techniques

MODULE 2: Neural Network

Introduction: Biological neurons, Models of single neurons, Articial and biological neural networks, Fundamental neurocomputing concepts, activation functions, neural network architectures, Single layer and Multi layers networks. Kohonen Self-Organizing Maps: The SOM algorithm

Training: Learning rules - Supervised learning and unsupervised learning. Hebb's rule, Delta rule, Multi-layer feedforward neural networks, Back propagation Learning.

MODULE 3: Fuzzy Logic Classical & Fuzzy Sets

Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions.

Fuzzy Logic System Components: Fuzzification, Membership value assignment, fuzzy IF-THEN rules and fuzzy reasoning, development of rule base and decision making system, fuzzy inference engines, defuzzification to crisp sets, defuzzification methods and their methods.

MODULE 4: Evolutionary Computation

Chromosomes, fitness functions, and selection mechanisms Procedure of Genetic Algorithm, Genetic Representation, Initialization and selection, Genetic Operators- cross over, mutation, Elite strategy and selection, Termination Criteria.

MODULE 5: Hybrid Intelligent Systems

Hybrid systems – integration of fuzzy logic, neural networks and genetic algorithms – nontraditional optimization techniques like ant colony optimization – Particle swarm optimization.

MODULE 6: Applications

Applications of ANN in load forecasting, detection of fault in distribution system, Power system dynamic load modelling, stability analysis.

Applications Fuzzy logic in Economic Dispatch, Unit Commitment, load flow study, detection of faults, Schedule maintenance, Electrical Machines.

Textbooks/ Reference Books:

1. Artificial Intelligence by Rich & Knight, Tata Mcgraw Hills
2. P. Jakson- Introduction to Expert Systems, Addison Wesley.
3. D.W. Ralston- Principles of Artificial Intelligence& Expert Systems, McGraw Hill.
4. George F.Luger, 'Artificial Intelligence – Structures and Strategies for Complex Problem Solving',FourthEdition,PearsonEducation,2002.
5. Janakiraman, K.Sarukesi, 'Foundations of Artificial Intelligence and Expert Systems', Macmillan Series in Computer Science.
6. W. Patterson, 'Introduction to Artificial Intelligence and Expert Systems', Prentice Hall of India, 2003.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE73	**Power System Instrumentation**	**3-0-0**	**3**

MODULE 1: Different Types of Instruments
Voltmeter –analog /digital, ammeter-analog/digital, frequency meter, induction type energy meter, 3 phase energy meter, Vector meter, tri vector meter, analog VAR meter.

MODULE 2: Measurement of Large Currents and Voltages
introduction, high voltage measurement technique for DC/AC voltage measurement, Buffer system: ratio and phase angle error, Pulse trained energy meter

MODULE 3: Measurements in Power Plants
Electrical measurements – Current, voltage, power, frequency, power factor etc. – Non electrical parameters – Flow of feed water, fuel, air and steam with correction factor for temperature – Steam pressure and steam temperature – Drum level measurement – Radiation detector – Smoke density measurement – Dust monitor

MODULE 4: Data Loggers and Data Display System

MODULE 5: Telemetry
Introduction, component, short term telemetering, Data transmission channels-PLCC, microwave links

MODULE 6: SCADA

MODULE 7: Introduction to PMUs and their Placement

Textbooks/ Reference Books:

1. Cooper Helfrick, “Electrical Instrumentation and Measuring Techniques”, Prentice Hall India, 1986
2. D. C. Nakra and K. K. Chowdhry, “Instrumentation, Measurement, and Analysis”, Tata McGraw Hill Publishing Co., 1984. 3. Selected topics from IEEE, AIEE and CIGRE Journals

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818OE41	**Reliability Engineering**	**3-0-0**	**3**

COURSE OBJECTIVES:
1. To study fundamental concepts of reliability and its applications in engineering
2. To study reliability evaluation techniques

COURSE OUTCOMES (COs):
After successful completion of the course student should be able to:
CO1: Understand fundamental concepts of probability and statistics and its applications in reliability engineering
CO2: Develop reliability block diagram and evaluate reliability of engineering systems.
CO3: Apply various techniques to evaluate reliability of components or systems using failure density functions and failure rate/hazard rate functions
CO4: Develop and analyze Markov model for a system and evaluate steady-state availability and reliability
CO5: Develop and implement maintenance based models for reliability evaluation of systems

MODULE 1: Introduction
Definition of reliability, reasons for reliability engineering programmes, applications and benefits, Failure- causes of failures, modes of failures, failure rate; life characteristics pattern (Bath-tub curve)

MODULE 2: Basic Probability Theory
Basic probability theorems, random variables, probability distributions, mean and standard deviation, median and mode, binomial distribution

MODULE 3: System Modeling-I
Reliability block diagrams: series, parallel, series-parallel combination, partially redundant system, standby system, complex systems; Reliability evaluation techniques

MODULE 4: System modeling-II
Failure density function, cumulative failure distribution function, reliability function, failure rate and hazard rate function, MTTF, MTTR and MTBF, Reliability evaluation using distributions

MODULE 5: Markov Modeling
Markov process and general concept of modeling; Markov models of repairable systems, Stochastic transitional probability matrix; Steady-state availability

MODULE 6: Maintained Systems
Maintenance: objectives, types and forms of maintenance, Maintainability; Preventive maintenance, Corrective maintenance; Maintenance based reliability models

Textbooks/ Reference Books:
1. Roy Billinton and Ronald N. Allan, Reliability Evaluation of Engineering Systems-concepts and techniques (2nd Edition), Plenum Press, 1992.
2. Singiresu S Rao: Reliability Engineering, Pearson Education, 2016
3. E. Balagurusamy: Reliability Engineering, Tata McGraw Hill Publishing Comp. Ltd., 1984.
4. E. E. Lewis, Introduction to Reliability Engineering, John Wiley and Sons, 1996.

5. A K. Govil: Reliability Engineering, Tata McGraw Hill Publishing Comp. Ltd., 1983.
6. Dimitri Kececioglu: Reliability Engineering Handbook (Vol 1), Prentice Hall PTR, 1991.
7. M. L. Shooman, Probabilistic Reliability- an engineering approach, McGraw Hill Book Company, 1968.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818OE42	**Web Technology**	**3-0-0**	**3**

MODULE 1: DHTML
Object model and Collections, Event model, Filters and Transitions, Data binding with tabular data control, XML – XML vocabularies, Document Object Model, SAX, Simple Object Access Protocol(SOAP), Extensible Style sheet Language(XSL)

MODULE 2: Server Side Scripting Languages
JSP- Introduction to JSP, JSP Architecture, Scripting components, Standard actions, JSP with JDBC.

MODULE 3: PHP
Introduction (variables, control statements etc.), String processing and regular expression, Form processing and business logic, connecting to a database, Cookies, Dynamic content in PHP

Text Books:

1. Deitel & Deitel, JAVA: How to Program, Pearson education, 7th edition, 2008.
2. Deitel & Deitel, Internet and World Wide Web How to Program, Pearson education, 3rd edition, 2005.

Reference Books:

1. Ivan Bay Ross, Web Enabled Commercial Application using Java 2, BPB publication, 1998.
2. David Flanagan, Java Script the Definitive Guide, O'reilly, 5th edition, 2006.

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Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818OE43	**Optical Communication Engineering**	**3-0-0**	**3**

COURSE OUTCOMES: At the end of the course the student will be able to:
CO1: Explain the properties of the optical fibers and optical components.
CO2: Analyze operation of lasers, LEDs and detectors
CO3: Analyze system performance of optical communication systems
CO4: Analyze non-linear effects in optical fibers

MODULE 1: Review of Theory of Light
Introduction to vector nature of light, propagation of light, propagation of light in a cylindrical dielectric rod, Ray model, wave model.

MODULE 2: Optical Fibers
Modal analysis of a step index fiber; Signal degradation on optical fiber due to dispersion and attenuation; Fabrication of fibers and measurement techniques like OTDR

MODULE 3: Optical Sources and Link Design
LEDs and Lasers, Photo-detectors - pin-diodes, APDs, detector responsivity, noise, optical receivers**;** Optical link design: BER calculation, quantum limit, power penalties.

MODULE 4: Optical Switches
Coupled mode analysis of directional couplers; electro-optic switches.

MODULE 5: Optical Amplifiers
EDFA, Raman amplifier; WDM and DWDM systems; Principles of WDM networks

MODULE 6: Nonlinear Effects in Fiber Optic Links
Concept of self-phase modulation, group velocity dispersion and solution based communication.

Text Books:

1. J. Keiser, Fibre Optic communication, McGraw-Hill, 5th Ed. 2013 (Indian Edition)

References:

1. T. Tamir, Integrated optics, (Topics in Applied Physics Vol.7), Springer-Verlag, 1975.
2. J. Gowar, Optical communication systems, Prentice Hall India, 1987.
3. S.E. Miller and A.G. Chynoweth, eds., Optical fibres telecommunications, Academic Press,1979.
4. G. Agrawal, Nonlinear fibre optics, Academic Press, 2nd Ed. 1994.
5. G. Agrawal, Fiber optic Communication Systems, John Wiley and sons, New York, 1997
6. F.C. Allard, Fiber Optics Handbook for engineers and scientists, McGraw Hill, New York(1990).

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818OE44	**Computer Graphics**	**3-0-0**	**3**

Course Outcomes (COs): At the end of this course students will be able to
CO1: Solve problems of Geometric Transformation applicable to Computer Graphics
CO2: Develop algorithms to draw specific objects on computer screen
CO3: Develop programs to draw specific objects on computer screen using OpenGL
CO4: Synthesize 3D objects for visualization on computer screen

MODULE 1: General Concepts
History of computer graphics, applications, graphics pipeline, physical and synthetic images, synthetic camera, modeling, animation, rendering, relation to computer vision and image processing, review of basic mathematical objects (points, vectors, matrix methods)

MODULE 2: Fundamentals of OpenGL
OpenGL architecture, primitives and attributes, simple modeling and rendering of two- and three-dimensional geometric objects, indexed and RGB color models, frame buffer, doublebuffering, GLUT, interaction, events and callbacks, picking.

MODULE 3: Geometric Transformations
Homogeneous coordinates, affine transformations (translation, rotation, scaling, shear), concatenation, matrix stacks and use of model view matrix in OpenGL for these operations.

MODULE 4: Viewing
Classical three dimensional viewing, computer viewing, specifying views, parallel and perspective projective transformations; Visibility- z-Buffer, BSP trees, Open-GL culling, hidden-surface algorithms.

MODULE 5: Shading
Light sources, illumination model, Gouraud and Phong shading for polygons, Rasterization- Line segment and polygon clipping, 3D clipping, scan conversion, polygonal fill, Bresenham's algorithm.

MODULE 6: Discrete Techniques
Texture mapping; compositing; textures in OpenGL; Ray Tracing- Recursive ray tracer, ray-sphere intersection

MODULE 7: Representation and Visualization
Bezier curves and surfaces, B-splines, visualization, interpolation, marching squares algorithm

Text Books:

1. Donald Hearn and Pauline Baker, Computer Graphics with OpenGL (third edition), PrenticeHall, 2003
2. Edward Angel, Interactive Computer Graphics. A Top-Down Approach Using OpenGL (fifthEdition), Pearson Education, 2008

References:

1. Computer Graphics (C Version) - D. Hearn and M. Pauline Baker, Pearson Education, 2nd Edition, 2004.

2. F. S. Hill Jr. and S. M. Kelley, Computer Graphics using OpenGL (3rd edition), Prentice Hall, 2006
3. Peter Shirley and Steve Marschner, Computer Graphics (first edition), A. K. Peters, 2010
4. Computer Graphics - Principles and Practice 2ndEd in C - J. D. Foley, A. Van Dam, S. K.Feiner and J. F. Hughes, Pearson Education, 2003.
5. Mathematical Elements for Computer Graphics, 2ndEd - D. F. Rogers and J. A. Adams,McGraw-Hill International Edition, 1990.
6. Procedural Elements for Computer Graphics, 2ndEd - D. F. Rogers and J. A. Adams,McGraw-Hill International Edition, 1990.
7. Computational Geometry Algorithms and Applications, 2ndEd – Mark de Berg, Marc van Kreveld, Mark Overmars and Otfried Schwarzkopf, Springer-Verlag, 2000.
8. Computational Geometry – Shamos Preparata
9. Geometric Data Structures for Computer Graphics – E. Langetepe and G. Zachmann, A K Peters©2006.

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